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Abstract Type: ○ Structured ◉ Unstructured

Unstructured Abstract*

Healthcare organizations face a "Triple Threat" of low analytics maturity, high workforce instability, and semant
together produce a crisis of "Institutional Amnesia." High leadership turnover, persistent digital health workforce
intent to leave among informatics specialists systematically erase the tacit knowledge required to navigate complex
trapping organizations in a cycle of low maturity where the rate of knowledge loss exceeds the rate of knowledge ca

Viewed through Nonaka's SECI model of knowledge creation, the root cause is a "Socialization Failure": high turno
networks required for mentorship, rendering the traditional apprenticeship model of informatics unsustainable. To a
a Design Science Research (DSR) approach, synthesizing evidence from healthcare informatics, knowledge management,
(2024-2026 workforce and NL2SQL literature) to develop a socio-technical framework called Human-in-the-Loop Knowled

HITL-KG is designed to shift the locus of organizational knowledge from volatile human memory to durable semantic
Query Triples," each comprising a natural language intent, executable SQL, and rationale metadata. By embedding kno
workflow of query generation, the framework aims to convert the ephemeral act of analytics into permanent instituti
Three-Pillar Assessment Rubric provides a structured self-assessment tool that enables organizations to identify co
across analytics maturity, workforce agility, and technical enablement.

A critical objection, the "Validator Paradox" (who validates the AI when experts leave?), is resolved by reframin
"Standard Work": each validated query establishes the current known standard rather than eternal truth, functioning
prevents regression. By decoupling analytical capability from individual tenure, healthcare systems can ensure that
even as the workforce evolves. This paper proposes and theoretically motivates the framework; empirical validation
study.

Indexing

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Keywords*

institutional amnesia, medical informatics, socio-technical systems, query governance, natural language processing, kr

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